PAWAR SHUBHAM

AI ENGINEER - LLM APPLICATIONS | COMPUTER VISION | MACHINE LEARNING |DOCUMENT AI

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CAREER PROFILE

AI Engineer with experience building and deploying production-oriented LLM applications, document-intelligence pipelines, retrieval-augmented generation (RAG) systems, computer-vision workflows, and AI-enabled business tools. Hands-on with Python, FastAPI, multimodal LLM APIs, OCR, vector retrieval, and containerized deployment. Published researcher in AI-based adaptive traffic control, with project funding and competition recognition.

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer | Persist Ventures | California – USA *Apr 2025 – Present*

- Build and maintain AI-powered web products using Python, FastAPI, React/Next.js, PostgreSQL, SQLAlchemy, Docker, AWS, and LLMs / LLM Apis.
- Built OCR and document-intelligence workflows supporting 7–10 document types using OpenAI Vision, PaddleOCR, OpenCV, and automated validation checks, achieving 93–97% field-level extraction accuracy on test documents and processing 30–40-page documents in approximately 3–5 minutes during observed usage.
- Developed an AI-powered PDF form-filling solution with 97% field-level content retrieval accuracy and 95–97% field-placement accuracy, using document-layout understanding and coordinate-based validation.
- Built Model Context Protocol (MCP) server with 8 tools to enable PDF form-filling workflows through Claude and OpenAI Codex.
- Integrated 6 government data sources, including Congress.gov, GovInfo, OpenStates, LegiScan, Regulations.gov, and SAM.gov, into legislative-intelligence pipelines processing 1,000–10,000 records with keyword and semantic search, structured extraction, LLM-generated summaries, weekly scheduled refreshes, and retry-with-backoff error handling.
- Developed an AI-powered content moderation system using self-hosted AI models deployed in containerized environments to automatically identify potentially inappropriate or policy-violating content.
- Integrated model inference into application workflows to enable automated content screening while reducing dependence on external AI APIs.
- Implemented RAG and code-assistance workflows using embeddings, document chunking, FAISS vector retrieval, semantic search, and prompt orchestration.
- Built AI-powered workflows for 4 business use cases: inventory extraction, recipe generation, pricing and profitability analysis, and content recommendations, using two-tower embeddings and FAISS retrieval.
- Developed computer-vision features for object detection, tracking, pose estimation, exercise recognition, and feedback using YOLO, MediaPipe, and OpenCV, achieving 77% mAP in an object-detection evaluation.
- Optimized multiple computer-vision models through INT8 quantization on AWS GPU-enabled EC2 instances, measuring 10–15% faster inference across multiple runs against FP32 baselines, with a 0–2 percentage-point mAP reduction.
- Improved backend responsiveness and processing workflows using asynchronous FastAPI endpoints, background jobs, WebSockets/SSE, caching, SQLAlchemy connection pooling, PostgreSQL indexing, query optimization, and schema improvements.
- Implemented application security using JWT authentication, OAuth2/PKCE, RBAC, endpoint-level permission checks, refresh-token handling, and password hashing.
- Configured CI/CD workflows covering automated builds and tests, Docker image publishing, Kubernetes deployments, environment secrets, and rollback procedures. Deployed applications across AWS, Kubernetes, staging, and self-hosted environments, with application logging, health checks, latency monitoring, AI usage monitoring, and automated failure alerts.

Python Developer Intern | CloudSpikes MultiCloud Solutions | Ontario – Canada *Nov 2024 – Dec 2024*

- Developed a PCB defect detection system using pretrained ResNet layers to extract deep visual features from printed circuit board images.
- Implemented embedding-based defect analysis by comparing feature vectors to identify visual differences and potential PCB defects.
- Applied transfer learning and feature similarity techniques to support automated PCB inspection and defect identification.
- Developed an AI-powered voice-calling workflow integrating knowledge-based preparation, speech-to-text, LLM-driven dialogue, text-to-speech, conversation memory, Flask APIs, and deployment components.
- Tested voice-calling workflows through actual phone calls, evaluating conversation flow and LLM response quality.
- Built Flask REST APIs with database-backed business logic and integrated services.
- Supported application deployment using Docker, Kubernetes, CI/CD pipelines, and Flask API deployment, with monitoring and logging.
- Containerized applications with Docker and supported Kubernetes deployment and CI/CD automation.

Python Intern | Infolabz | Gujarat – India *May 2021 – Jun 2021*

- Developed backend REST APIs and wrote automated unit tests using Python's unittest framework.
- Wrote 26–30 automated unit test cases using Python's unittest framework to validate API behavior and application logic.

AI PROJECTS

Dynamic Traffic Control System (DTCS) — AI-Based Traffic Management

Designed a vehicle-recognition and adaptive traffic-control system using live camera input and IoT hardware. Integrated image preprocessing and feature extraction, with emergency-vehicle prioritization through connected controllers. Research work published in a Springer journal; project received ₹90,000 in funding from the KSV SSIP Cell.

Tech: Python, TensorFlow, OpenCV, NumPy, Raspberry Pi, ESP32/ESP8266

Vehicle Counting and Speed Estimation

Built a video-based vehicle-counting and speed-estimation pipeline using YOLOv8 detection and centroid tracking. Estimated speed from frame rate and a reference distance, targeting real-time video processing.

Tech: Python, YOLOv8, Ultralytics, OpenCV, NumPy

SkyChords — AI Music Generator

Developed an AI-assisted melody-composition application with MIDI generation and audio-streaming features. The previous resume reports 90% accuracy for the melody-composition task.

Tech: Python, PyTorch, Flask, React Native

Phishy — Phishing Website Detection

Built a machine-learning classifier to identify phishing websites using extracted features. The previous resume reports 92% classification accuracy.

Tech: Python, scikit-learn, Random Forest, Decision Tree, Pandas

TECHNICAL STACK

Programming: Python, JavaScript, TypeScript, SQL

LLMs & Retrieval: OpenAI, Anthropic, Gemini, RAG, LangChain, LangGraph, prompt engineering, MCP, BERT

Computer Vision: OpenCV, YOLO, MediaPipe, PaddleOCR, OpenAI Vision

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, feature engineering, model evaluation, embeddings, FAISS, Pinecone

Backend & Data: FastAPI, Flask, REST APIs, WebSockets, SSE, PostgreSQL, SQLAlchemy, Alembic, Redis

Frontend: React, Next.js, React Native, HTML, CSS

Cloud & DevOps: AWS (EC2), Docker, Kubernetes, Argo CD, GitHub Actions, Nginx, Cloudflare

Security & IoT: JWT, OAuth 2.0, PKCE, RBAC, Raspberry Pi, ESP32, ESP8266, MicroPython

EDUCATION

Bachelors of Engineering in Computer Engineering *2022 – 2025*

LDRP Institute of Technology and Research **CGPA: 7.8**

Gandhinagar, Gujarat - India

Diploma in Computer Engineering *2019 – 2022*

Government Polytechnic – Ahmedabad **CGPA: 9.2**

Ahmedabad, Gujarat - India

ACHIEVEMENTS

- Research Publication: Served as the primary author of a research paper on AI-driven dynamic traffic control and emergency vehicle prioritization, published in Springer journal.
- Project Funding: Secured $1,000 in project funding from the SSIP Cell for the Dynamic Traffic Control System (DTCS).
- Project Achievement: Awarded 1st Place at Project Expo 2022, Government Polytechnic, Ahmedabad.